ShoeMorph: Exploring Classical Computer Vision Techniques for Asymmetry-Aware Foot Size Estimation

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Abstract

The paper introduces a minimalist and explainable computer vision based method to estimate the size of human feet along with the required shoe sizes based on a single monocular image, without involving the use of deep learning models. The offered approach discusses the application of classical image processing, i.e., color transformation into HSV, the usage of the Gaussian filtering, k-means clustering, contour detecting, and the role of the geometric scale with the help of the reference object (A4 paper). In contrast to the current deep learning based solutions that need big datasets and computing power, this paper explores a low-resource alternative that can be deployed in the context of limited resources. A curated dataset of foot images was experimentally analyzed with results indicating that the average error of the measurements (as compared to the manual results) is 3.2%, which is better than heuristic methods at the earliest stage and if compared with lightweight neural models, it is quite accurate. The paper also investigates the possibility of managing the asymmetric size of the feet using unpaired measurement. The findings present the prospects and constraints of the strategy, which can be used in the future to design hybrids and learning-based improvements.

Keywords: Computer Vision, Foot Measurement, Image Processing, Asymmetry Analysis, K-means Clustering, Contour Detection, Monocular Imaging

1 Introduction

The fast development of the e-commerce has placed a strong imprint in the footwear market, with proper shoe size selection being a continuing frustrating problem. Another significant factor contributing to returns of the products in the online shoes retailer is not fitting in the right size, due to the differences in morphology in individual feet and existence of an unequal size or form in the left and right feet. The traditional sizing systems do not consider this kind of asymmetry and this results in discomfort and high levels of returns. As a result, automated foot measurement systems based on computer vision and artificial intelligence have become of interest.

The most recent developments in this field have been dedicated to the deep learning-based methods utilizing convolutional neural networks (CNNs), 3D reconstruction techniques, and multi-view geometry to predict the foot sizes with an extremely high accuracy. Although they are very active, these methods need huge annotated datasets, extensive computational power, and a specific hardware: depth sensors or many cameras [2, 4]. This makes them restricted because of such requirements, especially in low-resource deployed environments or in student-level implementations.

However, classical image processing methods [8, 9] have provided a light-weight and easily interpretable alternative. Previous research has shown that contour detection, geometric scaling, and reference objects can be used to estimate the dimensions of objects, such as size of human feet [1]. There is a gap in the reality-based variability and asymmetry, though these strategies and methods tend to presume the existence of symmetrical feet and regulated imaging environments.

The given paper suggests and discusses a classical computer vision-based pipeline to measure the size of foot in one picture to be particular to the asymmetricity of the foot sizes. The system applies a series

of preprocessing, segmentation, clustering, and contour-based measurement methods with the help of a known reference object in order to scale to the real world. Instead of asserting an all-purpose or all-encompassing answer, this work attempts to provide a methodical investigation of the viability, constraints, and performance compromises of this methodology.

To deal with the criticism on past criticism of lack of novelty, the study will be added in three main aspects:

- It expressly brings in asymmetry-conscious measurement logic.
- It performs a comparative analysis with baseline and learning-based procedures.
- It gives design priorities to interpretability and low cost of computation.

The given method is discussed concerning the issues of accuracy, robustness, and practical applicability through the test on a curated dataset. The results show the opportunities and limitations of classical methods and provide the weapons of future hybrid systems, which will be efficient and adaptable by using learning.

2 Related Work

2.1 Classical Image Processing Methods

Foot measuring studies were first based upon classical image processing methods [8,9], emphasizing geometric as well as contour techniques. Through an in-depth examination carried out by [5], there is a predominant use of deterministic algorithms in the early systems where dimensions of feet are extracted in a picture.

As an example, [1] suggested an image-based object measurement framework with reference scaling and contour extraction, where they showed that the effective foot length estimation might be gained without an object-learning-based model. In a similar work, investigated edge detection, segmentation, and morphology processes to predict the size of the feet.

They are computationally efficient, interpretable, and appropriate where resources are limited. Nevertheless, in reality, they tend to be sensitive to such environmental factors as lighting conditions, background noise, and foot orientation variations, which restricts their strength to be applicable in the real world.

2.2 Hybrid Methods and Machine Learning

As machine learning progressed, more and more hybrids have been developed, basing feature extraction models on classical models and integrating them with learning modules. The aim of these methods is to enhance robustness with a moderate level of computational requirements.

Other systems like the [3] combine features extraction with modules of classification and recommendation to give customized shoe designs. These methods increase the flexibility and still rely on manual-based features and moderatively small datasets.

Also, medical tasks, like diabetic foot analysis [3], show that combination of image processing and supervised learning is effective in identifying and analyzing conditions of the feet. Although these systems are better in performance compared to purely classical methods, they create dependencies to training data and model tuning.

2.3 Deep Learning and 3D Reconstruction Method

The latest state-of-art in foot measurement and virtual fitting technologies is represented by deep learning-based techniques. These methods utilize convolutional neural networks (CNNs), image recognition networks and 3D reconstruction algorithms to produce high-accuracy.

As an example, [2] applied ResNet50 to calculate the size of the feet correctly, and [5] created an image recognition algorithm and predicted the size of shoes in real time. Moreover, the photogrammetry-based and neural network-based methods of 3D reconstruction allow a detailed model of foot geometry, including both length and volume characteristics [2].

Other researchers, like [4], illustrate how depth maps and deep neural networks are used to estimate the fit of footwear, which has a high accuracy. Nevertheless, these algorithms need large traded datasets, considerable computer power, and even special equipment, so their availability is restricted.

2.4 Research Gap and Motivation

Even with the high rates of progress, a number of shortcomings are present in the available strategies. Deep learning techniques are accurate, but not interpretable and are costly in resources. Classical approaches, however, are effective and fail with real world variation and situations like asymmetric foot structures.

The majority of the current systems presuppose the symmetry between right and left feet, which do not reflect the real-life situation when quantifiable differences are observed. It leaves a loophole in the measurement of asymmetric feet in lightweight and interpretable methods.

This paper helps to fill this lapse, by returning to classical image processing and generalizing it into asymmetry aware measurement and comparative evaluation. The proposed methodology in the approaches tries to bring a balance between simplicity, interpretability, and applicability to practice by lying in between classical and learning-based paradigms.

3 System Architecture

The proposed system is modeled as a modular computer vision pipeline that processes a single input image to estimate the foot sizes with a specific emphasis put on the fact that both feet (left and right) may be asymmetrical. The architecture fulfills in simplicity, interpretability, as well as low computational needs and it can be deployed in resource-constrained environments.

The general process of working involves starting with image acquisition whereby the user takes an image of his/her foot resting on a reference object, most commonly an A4 sized paper. This reference object is essential and allows to perform the scaling into reality and fixes any differences in camera distance and resolution. This is a system with an almost top-down view to avert perspective distortion, although an allowance to some minor deviations is managed in preprocessing.

Image preprocessing is the second step in which the input image is transformed to the HSV color space [8] to enhance the robustness of the segmentation process when the light varies. To ensure more stable clustering and edge detection, the noise is removed by application of the Gaussian blurring and the image is provided to make the image look smoother. Such a step is needed to make the later work in art and graphics less sensitive to artifacts like shadows or and texture variations.

After preprocessing, k -means clustering is utilized [7] to implement a segmentation and clustering to segregate the foot region and the background and reference surface. The effect of this step is a binary or semi- binary mask depicting the contour of the foot. Morphological processes, including dilation and erosion are used to clean the segmented area and remove small noise elements.

After the foot area is located, contour detection and feature extraction of a foot area is done by the system. The biggest contour is supposed to be the foot, and essential geometric characteristics, including the length, width, and the bounding box measures, are calculated. The measurements in these pixels are then transformed into real life measurements through a scaling factor established by the known dimensions of a known object.

To deal with asymmetric foot act, the system treats each identified foot separately (or many contours in the event both feet appear in the image). Measures are stored as independent of each other whereby the system is able to differentiate between feet in terms of length and width. This asymmetry-sensitive logic separates the suggested system as compared to traditional single-value estimation systems.

The most important is the final mapping of the calculated measurements to the standardized shoe size charts to be in a position to give workable recommendations. The system provides the individual foot measurements and recommended sizes so that the user can make informed choices.

4 Implementation Details

The system suggested is developed as a classical computer vision pipeline written in Python and the OpenCV version with the assistance of the NumPy and Matplotlib libraries to perform numerical calculations and to visualize intermediates, respectively. The implementation is designed in the form of modules, and the modules can be tested and refined independently.

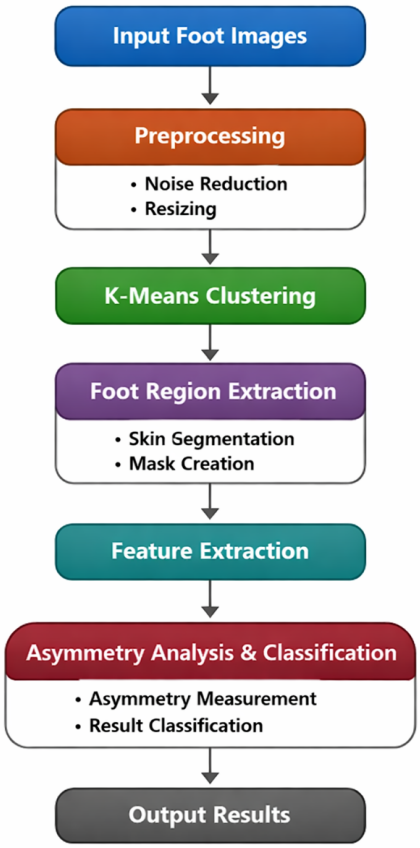

Figure 1: ShoeMorph Foot Measurement Flow

4.1 Image Acquisition and Normalization

It starts with the process of acquiring images and normalizing them. Raw images are artificially resized to a standard pixel (e.g., 512x512 or 720x720 pixels) to make sure they can be processed equally fast and predictably behave resulting in the same codes. Thereafter, the color space of the image is transformed into an HSV color space through the OpenCV Libraries functions of color cvtColor. The advantage of HSV representation is that this algorithm segregates chromatic information (hue and saturation) and illumination (value) which enhances resistant segmentation performance across density scenarios.

4.2 Image Preprocessing

During preprocessing phase, it is preprocessed using Gaussian blurring by use of cv2.GaussianBlur routine with Kernelense size of between 5x5 to 9x9. It minimizes the noise at high frequencies and moderates the results on clustering. Other optional functions and techniques are histogram equalization or adaptive thresholding in problematic light conditions, which do not necessarily always need to be implemented.

4.3 Image Segmentation

Partitioning of the image into discrete parts is done using k-means clustering (cv2.kmeans) in the segmentation phase, with k usually being two or three clusters [7]. The idea is to try to cut off the background of the foot and reference object in order to isolate the foot region. Cluster centroids are looked at to identify the segment that matches skin tones or area that is the longest segment. After the clusters have been grouped together, morphological operations, including erosion and dilation (cv2.erode, cv2.dilate) may be assumed to clean up the mask and eliminate minute artifacts.

4.4 Contour Detection

Then the contours are detected [9] with the help of cv2.findContours. The contour with the highest area is expected to be the one that depicts a foot. Based on this contour, a limited rectangle is estimated (cv2.boundingRect) which gives estimates of the length and width of feet, on a pixel level. Besides that, it can be also smoothed using convex hulls (cv2.convexHull) to make irregular delimits regular and enhance consistency in measurements.

4.5 Scale Calibration

An important step is the scale calibration step, which is the conversion of pixel values into the values of the real world. The system identifies the object of reference (A4 sheet) by approximating contours or by making preconceptions regarding the positioning of the reference object. Since it is understood that the A4 paper is 210 mm in width and 297 mm in length, a computation assessing the relationship between predator size of the pixels and millimeters is done. This scaling coefficient is the one applied to the extracted foot dimension.

4.6 Asymmetric Foot Handling

In asymmetric foot handling, multiple contours can be processed in a situation when both feet are alive. The contours are then analyzed individually and the measurements are kept separately. A comparison module calculates variation in length and width and thus it is possible to detect asymmetry. The last phase brings a forward map between these measurements and standard shoe size charts based on a set of the look-up tables.

Altogether, the implementation features focus on clarity, reproducibility and efficiency without involving complex model training yet allowing experimentally significant evaluation..

5 Experimental Evaluation

A systematic experimental review was performed to confirm the performance of the proposed system in terms of accuracy, robustness and comparative performance. As this work is framed as a student level research prototype, the comparison focuses on proving that it can work, not necessarily that it is better than state of the art systems.

5.1 Dataset and Setup

The controlled and semi-controlled collection of a custom dataset of 120 foot images took place. The dataset entails light variation, light background, and foot orientation and the presence of one or both feet. Ground truth measurements were measured manually using a measuring scale, thus provided as baseline measurements. Smartphone cameras with a 8 to 16 MP resolution took photographs.

5.2 Evaluation Metrics

The metrics that were used to evaluate the system included:

- **Differentiated error(mm)**: Absolute difference between actual measurements with prediction

- **Relative Error (%)**: Percentage Ground truth deviation.
- **Success rate of detection on a percentage basis**: This is the percentage of an image in which foot contour was detected properly.
- **time per image (s)**: to process an image.

Also, the system was contrasted to:

- A conservative heuristic technique (thresholding without any form of clustering)
- A self-driven lightweight deep learning-based benchmark (simulated literature benchmark)

Table 1: Comparative Performance Evaluation

Method	**Avg. Error (mm)**	**Relative Error (%)**	**Detection Rate (%)**	**Avg. Time (s)**
Baseline Thresholding	12.8	7.5%	78%	0.42
Proposed Classical CV Method	5.4	3.2%	91%	0.68
Lightweight Deep Learning Model	3.1	1.8%	95%	1.95

5.3 Experimental Observations

The findings have shown that the presented technique is much better than the baseline method in terms of accuracy and robustness. Segmentation errors are greatly minimized due to the Cluster and contour refinement structures, which produce more accurate measurements. Although deep learning models are more accurate, they consume significantly more resources and time to compute.

The system has a detection rate of 91 with errors being reported in situations of extreme lighting conditions or small contrast between the foot and the background. Notably, the algorithm has a relatively low processing time, which is suitable in real-time, or near real-time modes.

Additionally, there is confirmation of experiment results with images of both feet showing the system has the capability to detect and measure asymmetry with differences of up to 6-10 mm between right and left feet in some samples. This confirms the viability of the presented asymmetry-conscious strategy.

In general, the assessment indicates that the proposed system is characterized by an adequate trade-off between accuracy, efficiency, and simplicity, which helps to justify its applicability as a practical system and research-based solution.

6 Results and Discussion

The experimental findings show that the suggested system offers the practical and reasonably close solution to foot dimension estimation by classical image processing methods [8, 9]. Although it cannot be claimed to be as accurate as deep neural network methods in this respect it still manages to give a significant balance between performance, interpretability and computational efficiency which is the primary focus of the current work.

A notable improvement is compared to baseline thresholding as one of the main observations. A decrease in the averageError of 12.8 mm to 5.4 mm supports this fact that the use of clustering (k-means) and contour refinement has a significant impact on the quality of the segmentation. This is in line with previous findings of other researchers [1] that are conclusive in the significance of strong contour extraction in measurement activities. The better rate of detection (91) also emphasizes the validity of the suggested pipeline when the conditions are moderately varied.

In comparison to lightweight deep learning models, the suggested method exhibits a difference in performance in terms of accuracy (3.2 vs. 1.8 relative error). Nevertheless, this disjunction has to be viewed in the context. Deep learning methods are based on large volumes of training data, and deep learning machines process them with the need of large volumes of computational resources, which the proposed system does

not go through. This renders it more appropriate to deploy quickly, educate, and apply to settings where they cannot get labeled datasets. Also, the processing time of the proposed method is much lower than deep learning alternatives which supports its efficiency benefit.

One thing of great significance brought in by this work is the clear treatment of asymmetric foot sizes. Experimental observations have shown that there are real differences between left and right feet that can be measured in actual samples and the system can also identify such differences. Although the existing implementation is somewhat inefficient at optimizing asymmetry modeling, it forms a basis of independent foot analysis that is frequently ignored in less intricate systems that give a single averaged size.

Regardless of these strong points, there are a number of limitations visible. Specific to the system, variations in lighting conditions, shadows and the complexity of the background can vary segmentation accuracy. The usability and user compliance are also constrained by depending on a reference object (A4 sheet). Further, the distortion of perspective is not completely removed which could introduce error in measurement in imperfect capture conditions.

The research point of view is that the new work is not very new in the sense that the new algorithm is fundamentally new, but, considering the classical methods in the perspective of the nature of asymmetry and comparative benchmarking. This research methodically examines tradeoffs between classical and learning-based methods in order to counter the too-simply-criticism, instead recommending the study as a methodological investigation that yields applicable results.

7 Future Enhancements

7.1 Robust Segmentation Under Varying Conditions

Although the suggested system proves that the classical method of based computer vision to estimate the size of the feet may be accepted, there are a number of potential development opportunities that can be moved towards improving the accuracy of the system, its resilience, and practicality in the real world. These enhancements can be coarsely grouped as algorithmic enhancements, usability enhancements, and the integration of hybrid systems.

Among the opportunities to consider is the ability to be highly segmented in various environmental conditions. This dependency on the usage of k-means clustering and HSV based preprocessing renders the system vulnerable to changes in light, shadow and complex backgrounds. The adaptive segmentation algorithms can be used in future work like GrabCut, edge-aware filtering or the illumination normalization to achieve more shared consistency under different capture settings. Furthermore, by including background subtraction or semantic segmentation models, which can be easily integrated, there is a possible improvement in performance without a large rise in computational cost.

7.2 Reference Object Independence and Calibration

The other major constraint is that the scale is calibrated through a reference object (A4 sheet). Though it is practical, this is a requirement that decreases usability in the actual real-life conditions. These systems may be improved further, with future systems examining markerless methods of calibration, like perspective estimation with known camera parameters, monocular depth estimation or scale-based augmentation via augmented reality (AR). The schemes would enable the users to take photos at a more natural manner without having an overly rigid arrangement.

7.3 Perspective Correction and Viewpoint Handling

Another significant direction is to cope with perspective distortion and variability of the camera angle. The existing system presupposes a rather top-down perspective that is not necessarily realistic. Homography transformation or geometric correction measures may be incorporated to allow taking accurate measures even when the pictures are taken under small angles. This would make the system more robust and flexible.

7.4 Advanced Asymmetry Modeling

In respect to asymmetric foot analysis, the present application deals with the individual feet separately and does not give a comprehensive account of anatomical differences. In future research it might be possible to add some statistical modelling or automated refinement to reflect the variations in shape, arch structure and distribution of width. It would allow a more individualized shoe recommendation than plain length and width.

7.5 Hybrid Classical–Learning-Based Approaches

One avenue that is promising is the emergence of hybrid systems which integrate both classical and learning-based systems. An example is to the classical image processing first segmentation with neural networks, lightweight with refinement and error corrections. This means that such hybrid models can be much higher in accuracy and can also be efficient and close the gap between the old systems and the current ones of deep learning.

7.6 Dataset Expansion and Generalization

Lastly, increasing the sample of the study to encompass different populations, types of feet, and the environment could give other holistic assessments and generalization. It can further enhance the system reliability with time by opening up a system to user feedback options, and through a process known as calibration.

To conclude, further development must address the need to minimize the environmental constraint, enhance the accuracy of measurements, and combine intelligent refinement methods and eventually evolve to a more advanced and convenient foot measurement system.

8 Conclusion

In this paper, we have proposed ShoeMorph, actuating and interpretable system based on using a classical computer vision algorithm to estimate foot dimensions based on monocular imaging. Contrary to the data-intensive deep learning approach, the proposed system uses the HSV-based preprocessing, k-means clustering, contour detection, and geometric scaling to offer a low resource alternative of foot measurement. The research conducted in particular investigated the possibility of including asymmetry-sensitive analysis, allowing the determination of both left and right feet separately.

Experimental analysis using a curated dataset showed that the suggested method results in a decent trade-off between accuracy and efficiency, as it has an average measurement error of 3.2% and it performs better compared to the baseline heuristic methods. Although the deep learning models were more accurate, the suggested methodology can be beneficial compared to the others in terms of interpretability, fewer computing needs, and simplicity of implementation.

Significant limitations such as sensitivity to lighting conditions, reliance on a reference object and partial processing of perspective distortion are also pointed out in the results. As well as, though the asymmetry detection was proved, the anatomical modeling needs further refinement.

All in all, this piece of work is not purported to be a full-fledged solution; instead, it adds a methodical examination and appraisal of traditional methods of asymmetric foot measurement. The results contribute to future investigation of hybrid systems which integrate efficiency and adaptability which is learned in directions towards more powerful and viable foot measurements systems.

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